

```
1 import java.util.Scanner;
2
3 public class GradeAnalysisSystem {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8         int classNum;
9
10        System.out.println("=====");
11    );
12        System.out.println("                GRADE ANALYSIS SYSTEM");
13
14        System.out.println("=====\\
15n");
16
17        // Number of Students
18        while (true) {
19            System.out.print("Enter number of students: ");
20
21            if (!sc.hasNextInt()) {
22                System.out.println("Invalid Input. Please enter a number.");
23                sc.next();
24                continue;
25            }
26
27            classNum = sc.nextInt();
28
29            if (classNum < 2) {
30                System.out.println("Number of students should not be less than 2.");
31                continue;
32            }
33
34            sc.nextLine();
35
36            System.out.println("_____\\
37n");
38            break;
39        }
40
41        // Student Names
42        String[] student = new String[classNum];
43
44        for (int i = 0; i < classNum; i++) {
45            while (true) {
46                System.out.print("Enter Student Name " + (i + 1) + " (LAST NAME,
47FIRST NAME): ");
48                student[i] = sc.nextLine();
49
50                if (student[i].matches("[A-Z ]+\\s*[A-Z ]+")) {
51                    break;
52                }
53
54                System.out.println("Invalid format. Please use: LAST NAME, FIRST
55NAME");
56            }
57        }
58
59        // Student Grades
```

```
53     int[] grades = new int[classNum];
54
55     System.out.println("\n_____
-");
56
57     for (int i = 0; i < classNum; i++) {
58         while (true) {
59             System.out.print("Enter grade of " + student[i] + ": ");
60
61             if (!sc.hasNextInt()) {
62                 System.out.println("Invalid input. Please enter a number.");
63                 sc.next();
64                 continue;
65             }
66
67             grades[i] = sc.nextInt();
68
69             if (grades[i] < 0 || grades[i] > 100) {
70                 System.out.println("Grade must be between 0 and 100.");
71                 continue;
72             }
73
74             break;
75         }
76     }
77
78     // Display ranking and highest performer(s)
79     topPerforming(student, grades);
80
81     // Display class average
82     classAverage(grades);
83
84     // Display number of students who passed and failed
85     studentsPassFail(grades);
86 }
87
88 // Class list and highest performing student(s)
89 public static void topPerforming(String[] student, int[] grades) {
90
91     // Sort from highest to lowest
92     for (int i = 0; i < grades.length - 1; i++) {
93         for (int j = i + 1; j < grades.length; j++) {
94
95             if (grades[j] > grades[i]) {
96
97                 // Swap grades
98                 int tempGrade = grades[i];
99                 grades[i] = grades[j];
100                 grades[j] = tempGrade;
101
102                 // Swap student names
103                 String tempStudent = student[i];
104                 student[i] = student[j];
105                 student[j] = tempStudent;
106             }
107         }
108     }
109 }
```

```

110 System.out.println("\n=====
=");
111 System.out.println("                                CLASS RANKING");
112 System.out.println("=====
");
113
114 System.out.printf("%-5s %-30s %-7s %-8s %-8s\n",
115     "Rank", "Student Name", "Grade", "Letter", "Status");
116
117 System.out.println("-----
-----");
118
119 for (int i = 0; i < grades.length; i++) {
120     String status = (grades[i] >= 60) ? "Passed" : "Failed";
121
122     System.out.printf("%-5d %-30s %-7d %-8s %-8s\n",
123         i + 1,
124         student[i],
125         grades[i],
126         getLetterGrade(grades[i]),
127         status);
128 }
129
130 System.out.println("=====
");
131
132 System.out.println("\nTop Performing Student(s):");
133
134 int highest = grades[0];
135
136 for (int i = 0; i < grades.length; i++) {
137     if (grades[i] == highest) {
138         System.out.println(student[i]
139             + " - " + grades[i]
140             + " (" + getLetterGrade(grades[i]) + ")");
141     } else {
142         break;
143     }
144 }
145 }
146
147 // Calculates class average
148 public static double classAverage(int[] grades) {
149
150     int sum = 0;
151
152     // Compute the sum of all grades
153     for (int i = 0; i < grades.length; i++) {
154         sum += grades[i];
155     }
156
157     double average = (double) sum / grades.length;
158
159     boolean belowSeventy = average < 70;
160
161     System.out.println("\n=====

```

```
=");
162     System.out.println("                                CLASS AVERAGE");
163
    System.out.println("=====");
);
164     System.out.printf("Class Average: %.2f out of 100%n", average);
165     System.out.println("Did the class average fall below 70? " + belowSeventy);
166
167     return average;
168 }
169
// Counts the number of passed and failed students
170 public static void studentsPassFail(int[] grades) {
171
172     int passedCtr = 0;
173     int failedCtr = 0;
174
175     for (int i = 0; i < grades.length; i++) {
176
177         if (grades[i] >= 60) {
178             passedCtr++;
179         } else {
180             failedCtr++;
181         }
182     }
183 }
184
185
    System.out.println("\n=====");
=");
186     System.out.println("                                PASSED / FAILED");
187
    System.out.println("=====");
);
188     System.out.println("Students Passed : " + passedCtr);
189     System.out.println("Students Failed : " + failedCtr);
190 }
191
// Calculates letter grade of student
192 public static String getLetterGrade(int grade) {
193
194     if (grade >= 98)
195         return "A+";
196     else if (grade >= 92)
197         return "A";
198     else if (grade >= 87)
199         return "B+";
200     else if (grade >= 81)
201         return "B";
202     else if (grade >= 77)
203         return "C+";
204     else if (grade >= 71)
205         return "C";
206     else if (grade >= 60)
207         return "D";
208     else
209         return "F";
210 }
211 }
212 }
```